This highlights an important aspect about Bayesian learning: our prior determines how learning is done.

- Bob's prior is fairly high, so by the end of the one hundred tests, he's completely convinced that the person is psychic.
- We, on the other hand, are more skeptical in that our initial prior is much lower than Bob's. This means it would take more evidence for us to arrive at a high posterior.
- Alice's complete disregard of the possibility, denoted by her zero prior, prevents her posterior probability from changing from zero.

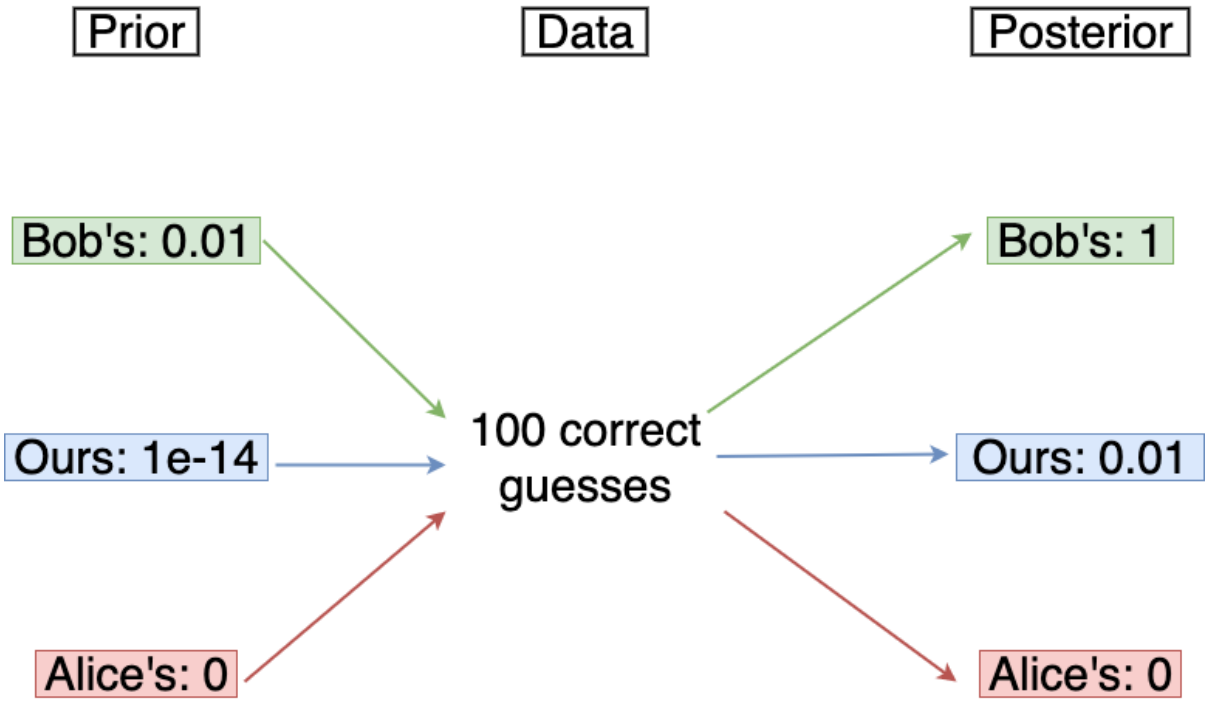


Figure 3.3 How each person's prior belief is updated by the same data. Comparing to Bob's, our prior is much lower and increases more slowly. Alice's prior is 0 and stays at 0 throughout.

This is summarized by Figure 3.3. Although the claim in our example is about the event that someone is a psychic, the same Bayesian update procedure applies to all situations where we have a probabilistic belief about some event and frequently update it in light of data. In fact, this is actually the reason why sometimes we can't seem to be able to change someone's mind, even in the face of overwhelming evidence: because they start out with zero prior probability, and nothing will update the posterior to anything other than zero.

This discussion is philosophically interesting, in that it shows in order to be able to convince someone of something, they need to at least entertain the idea by assigning non-zero prior probability to that event. More relevant to our topic, the example shows the importance of having good prior knowledge for a Bayesian model. As we have said, we specify a GP's prior knowledge with the mean and the covariance functions. Each choice leads to a different behavior in the GP's prediction.

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